Lesson 5: Introducing Encapsulation and Data Validation

Focus: Protecting object attributes and controlling how data is accessed or modified.

The school has discovered that some users are entering invalid marks, such as 120 or -15. The system must prevent this.

Exercise 5 – Protecting Student Information

Continue with the existing `Student` class.

Questions

1. Modify the class so that the student's marks cannot be accessed or modified directly from outside the class. Use private attributes or the equivalent access-control mechanism in your programming language.
2. Create a method called `setMathsMark(newMark)` that changes the Mathematics mark.
3. The method must only accept marks between 0 and 100.
4. If an invalid mark is entered, display an appropriate error message and retain the previous mark.
5. Create similar methods for Computer Science and Physics marks.
6. Create methods that allow users to retrieve each subject mark.
7. Test the class by attempting to:
    - Enter a valid mark.
    - Enter a negative mark.
    - Enter a mark greater than 100.
    - Retrieve a student's mark.

Challenge questions

8. Add validation to prevent a student's age from being less than 14.
9. Prevent a student's name and surname from being left blank.
10. Explain why encapsulation is important when developing a student management system.

Learning checkpoint: Learners should understand encapsulation, access control, getters, setters and validation.